Literature Review: Machine Learning for Scientists

Nicholas Hurley, Ashley Yuan, Sara Maria Langner

Topic 1: Machine Learning x Phases in Condensed Matter Systems

Sara Maria Langner

Paper 1: Carleo & Troyer — Neural quantum states

This paper aims at reducing complexity of the many-body wave function by using artificial neural networks characterized by a set of parameters W the NN can be trained on to accurately find the ground state and time-dependent states for any given Hamiltonian H. So far this has been applied to 1D and 2D fermionic systems. The paper presents results for the antiferromagnetic Heisenberg model in 1D and 2D as well as the Ising model in 1D. As a follow up experiment one could use the method presented to determine many-body wavefunctions for a 2D Ising model. Additionally, these methods should be generalized in the future such that they can be used for arbitrary many-body systems in any dimension.

Paper 2: Carrasquilla & Melko — Machine learning phases of matter

The authors discuss performing supervised machine learning using neural networks to classify phase transition by order parameters. One could extend these methods to systems that do not only have one but multiple order parameters.

Paper 3: van Nieuwenburg et. al. — Learning phase transitions by confusion

Here, phase transitions in condensed matter systems are identified by combining supervised and unsupervised machine learning methods. One of the main benefits of this method is that generalization is quite accessible. However, I'm wondering what else is advantageous about this approach especially compared to results obtained with both, supervised and unsupervised, methods respectively. As mentioned in the paper, one could perform the presented technique on systems with more than one phase transition to see how that changes the shape of the performance function. As mentioned above, I am missing quantitative results showing the advantage of the confusion scheme compared to others. Also it only answers the where about phase transitions and not the why which is, in my opinion, one of the more harder and crucial problems to solve.

Topic 2: Machine Learning in Quantum Control, Dynamics, and VQEs

Ashley Yuan, Nicholas Hurley

Paper 4: Fast and Noise-aware Machine Learning Variational Quantum Eigensolver

This paper looks at the classical optimization bottleneck inherent in VQEs. Instead of relying on traditional optimizers that require many unpredictable iterations, the authors train a feedforward deep neural network on intermediate parameter data and measurements to directly predict

optimal final parameters. They tested this method on real IBM superconducting quantum devices to find the ground state energies of molecules like H2, H3, and HeH+. What is particularly surprising is that the intermediate data contains enough information about the device's hardware noise that the machine learning model can natively compensate for arbitrary coherent noise across different noisy processors. As a followup, you could ask whether this neural network approach can be generalized to predict parameters across entirely different molecular Hamiltonians without needing to be retrained from scratch for each specific molecule.

Paper 5: Simulating and Learning Quantum Evolution: A CTQW-ML Framework
This paper explores using data driven techniques to simulate the time dependent Schrödinger equation. The authors use Continuous-Time Quantum Walks (CTQW) to apply unitary evolution and generate probability amplitude distributions. They then implement a supervised neural network model to learn and predict the squared modulus of the wavefunction based on both spatial and temporal coordinates. It is impressive how accurately the machine learning model reproduces the qualitative structure and temporal progression of the quantum system, such as wavepacket oscillation and revival. A crucial next question would be how efficiently this CTQW-ML framework scales when applied to multi-qubit systems, since the exponential growth of the Hilbert space is the traditional computational bottleneck for propagating Schrödinger dynamics.

Paper 6: Machine Learning-Assisted Manipulation and Readout of Molecular Spin Qubits
This work is at the intersection of AI and quantum control by applying Artificial Neural Networks (ANNs) to assist in the resonant readout and amplitude/phase manipulation of qubits. While the experimental demonstration uses a molecular spin qubit, the authors specifically note that the method is generalizable to systems showing quantum coherence via echo signals, including superconducting qubits. What stands out is how the ANNs process refocusing echo signals to improve state discrimination—readout being one of the most error-prone operations on quantum processors. Building on this, it would be interesting to ask how the latency of this neural-network-assisted readout compares to traditional FPGA-based discrimination, and whether it could be integrated into a real-time feedback loop alongside quantum optimal control algorithms (like Piccolo or GRAPE).

Paper 7: Direct Collocation for Quantum Optimal Control

What they did? In this paper, researchers adapted typical gradient-based methods for solving quantum optimal control problems. Previous **indirect methods** treated a control sequence's final state/unitary as the sole objective. This is indirect because the drive-controls are the only decision variables, and the intermediate states are not considered by the optimizer. In this paper, the intermediate states are treated as **direct** decision variables alongside the drive-control values. Feasibility is an added constraint, which enforces that unitary rollout between time-steps is self-consistent. By treating the intermediate states as decision variables, the optimizer can violate physically realistic dynamics en route to the final solution. Another

contribution was in their use of the **Pade Integrator**, which quickly approximates matrix-exponentiation, allowing for more efficient unitary rollout during optimization.

What surprised you? The authors additionally treated time-step durations as decision variables, penalizing longer durations. This allowed them to optimize **minimum-time** solutions. Indirect methods prohibited such problem landscapes, because shrinking time-steps would naturally raise the objective if proper dynamics were necessitated.

What would you ask next? How can this be applied to non-unitary systems, utilizing the Lindblad master equation to simulate thermal decoherence? What other decision variables can we include in our objective (e.g. energy, entropy)? How is this applied to perturbative Hamiltonians, and how can we shape the objective to stay in the perturbative regime? How can this be enhanced/targeted for adiabatic systems?

What were you confused about? How is constraint enforcement using Lagrange multipliers (IPOPT methods) fundamentally different than unitary/state fidelity metrics?

Paper 8*:* Provably Efficient Self-Calibrating Quantum Fault Tolerance

What they did? The authors showed that the optimal operating point for quantum error-correcting (QEC) controls can be optimized in real time. Since well-calibrated QEC can now keep logical qubit error gates below the necessary threshold for fault-tolerant quantum computing, one next needs to consider realistic noise in control instruments, as well as environmental drifts of the optimal calibration point. The authors showed that the rate of detector-events in the logical qubits forms a locally-convex cost landscape, and drives can therefore be optimized to minimize these detector events after each batch of syndrome measurements. They showed that this is done for one-time control drifts, as well as for drifts in the optimal operating point of the qubit. They modeled this numerically for pulse-level Rydberg-atom simulations

What surprised you? The authors were able not only to show that online self-calibration was possible for drifting controls, but also for drifting operating points. These corrections are zeroth order, and without being able to extract gradient information, I am surprised that they were able to do both.

What would you ask next? How does one detect spurious control noise? How can one distinguish some Poissonian distribution of error from a continuously drifting control parameter (i.e. magnitude, phase, frequency)? Is there any other metric we can use to extract gradient information about this cost landscape, so that we can pursue first-order, more efficient optimizations?

What were you confused about? I was first confused on how drifting operating points are mathematically distinct from drifting controls, and how/whether they can be corrected for simultaneously. I am also confused how gradients can be estimated using SPSA using only two measurements.

Paper 9: Reinforcement Learning for Quantum Control under Physical Constraints

What they did? The authors used reinforcement learning to optimize control sequences for generating quantum states. As opposed to gradient-methods like PICO and GRAPE, which utilize Hamiltonian gradients/Hessians, this method used reinforcement learning to optimize a pulse. They trained a neural network policy, which outputs the Gaussian distribution parameters of a time-dependent control vector. By sampling the controls from this prior distribution and computing its generated fidelity, they generated a clipped surrogate loss through Proximal Policy Optimization, which then allowed them to update the policy. By continuously updating their policy through this method, they were able to converge upon high fidelity pulses. They also used physics constraints such as smoothness and drive amplitude to enforce sufficient adiabaticity and smooth rollouts.

What surprised you? I am surprised that they were able to converge upon optimal solutions without any knowledge of the Hamiltonian dynamics, which would typically be necessary for computing gradients/curvature.

What would you ask next? I am curious how this could be improved with a hybrid-approach, which blends *approximate* knowledge of Hamiltonian parameters (e.g. couplings, frequencies) with unoptimized controls. Could this be used to determine the Hamiltonian parameters while also optimizing the control pulses, given some prior?

What were you confused about? The state/context of the policy network seemed irrelevant to the problem. What does this input represent? What degrees of freedom does it offer? I am still unclear on how/whether the approximate curvature of the loss landscape is determined merely through fidelity calculations.